Algebraic and Differential Topology – Lecture Notes

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1 Introduction

Definition 1.1.
A topology on a set X is a collection τ of subsets of X such that:

(i) $\varnothing, X \in \tau$.

(ii) If $U_i \in \tau$ for all $i \in I$, then $\bigcup_{i\in I} U_i \in \tau$.

(iii) If $U_i \in \tau$ for all $i \in I$ where I is finite, then $\bigcap_{i\in I} U_i \in \tau$.

We call the pair (X, τ) a topological space. The elements of τ are called open sets.

Definition 1.2.
A **basis** for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

(i) $\forall x \in X$, $\exists B \in \mathcal{B}$ containing x.

(ii) If $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$, then there is a basis element $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

$\mathcal{B}$ generates a topology τ on X as follows:
$U \in \tau$ if $\forall x \in U$, $\exists B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark 1.3.
It is important that $B \in \mathcal{B}$ is a subset of U in the last statement. If we only required $x \in B \cap U$, we would run into problems. For example consider $X = \mathbb{R}$ and $U = \{0, 1\}$. We know that $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a < b\}$ is a basis for the standard topology on $\mathbb{R}$. Then for $x = 0$, we can choose $B = (-1, 1)$ and for $x = 1$, we can choose $B = (0, 2)$. This would mean that U is open, which should not be the case.

Example 1.4.

- (M, d) is a metric space. Then $\mathcal{B} = \{B_r(x) : x \in M, r > 0\}$ is a basis for the metric topology on M, where $B_r(x) = \{y \in M : d(x, y) < r\}$ is the open ball of radius r around x.
- X, Y topological spaces. The product topology on $X \times Y$ is generated by the basis $\mathcal{B} = \{U \times V : U \subseteq X, V \subseteq Y \text{ open}\}$.
- subspace topology: Let (X, τ) be a topological space, $S \subseteq X$. Then the subspace topology of S is $\tau_S = \{U \cap S : U \in \tau\}$.

Definition 1.5.
U is a neighbourhood of x if $\exists V$ open such that $x \in V \subseteq U$.

Definition 1.6.
Let X, Y be topological spaces.
A map $f : X \to Y$ is continuous if $\forall U \subseteq Y$ open, $f^{-1}(U)$ is open in X.
If f is a bijection and both f and f^{-1} are continuous, then f is called a homeomorphism. In this case, we say that X and Y are homeomorphic or topologically equivalent ($X \simeq Y$).

Example 1.7.

- $f : [0,1] \to \mathbb{S}^1$ given by $f(t) = (cos(2\pi t), sin(2\pi t))$.
  f is continuous and bijective but f^{-1} is not continuous.
  $f([0,1))$ is not open in $\mathbb{S}^1$.
- $\mathbb{R} \not\simeq \mathbb{R}^2$: Assume they are homeomorphic. Then there exists a homeomorphism $f : \mathbb{R} \to \mathbb{R}^2$ with $f(0) = (a,b)$. But $f|_{\mathbb{R}\setminus\{0\}} : \mathbb{R} \setminus \{0\} \to \mathbb{R}^2 \setminus \{(a,b)\}$ is a homeomorphism. But $\mathbb{R} \setminus \{0\}$ is not connected, while $\mathbb{R}^2 \setminus \{(a,b)\}$ is connected. This is a contradiction.
- $\mathbb{R}^2 \not\simeq \mathbb{R}^3$: Assume they are diffeomorphic. Then there exists a diffeomorphism $f : \mathbb{R}^2 \to \mathbb{R}^3$. Then $f \circ f^{-1} = id_{\mathbb{R}^3}, f^{-1} \circ f = id_{\mathbb{R}^2}$ so $Df_x Df_y^{-1} = I_3$ and $Df_y^{-1} Df_x = I_2$.

2 Homotopy

Definition 2.1.
Let X, Y be topological spaces (as always in this chapter) and $f, g : X \to Y$ be continuous maps. Then f is called homotopic to g if there exists a $F : X \times I \to Y$ continuous s.t. $F(x,0) = f(x)$ and $F(x,1) = g(x)$ for all $x \in X$. We write $f \simeq_F g$ and F is called a homotopy between f and g. If $f \simeq c$ where c is a constant map, then f is called null-homotopic.

Definition 2.2.
A continuous map $f : I \to X$ is called a path in X.
Two paths $f, g : I \to X$ are called **path homotopic** if they have the same start and end points, i.e. $f(0) = g(0)$ and $f(1) = g(1)$, and there exists a continuous map $F : I \times I \to X$ s.t. $F(x,0) = f(x)$, $F(x,1) = g(x)$ for all $x \in I$, and $F(0,t) = f(0) = g(0)$, $F(1,t) = f(1) = g(1)$ for all $t \in I$. We write $f \simeq_F g$.

Remark 2.3.
This is an equivalence relation:

(i) $f \simeq_F f$ with $F(x,t) = f(x)$ for all $x, t \in I$.

(ii) $f \simeq_F g \implies g \simeq_G f$ with $G(x,t) := F(x, 1-t)$.

(iii) $f \simeq_F g$, $g \simeq_G h \implies f \simeq_H h$ with $\begin{cases} H(x,t) = F(x,2t) & 0 \le t \le \frac{1}{2} \\ H(x,t) = G(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}$

If f is a path then we call $[f]$ the path homotopy equivalence class of f.

Example 2.4.

- Any two continuous maps $f, g : X \to \mathbb{R}^2$ are homotopic with $F(x,t) = (1-t)f(x) + tg(x)$.
- Let $X = \mathbb{R}^2 \setminus \{(0,0)\}$ and let $f, g : I \to X$ be defined by $f(t) = e^{\pi i t}$ and $g(t) = e^{-\pi i t}$. Then f and g are not path homotopic in X. Proving this is not trivial. First, we state an equivalent problem which is slightly easier to prove. Instead of two paths, we try to show that the unit circle as a path $h(t) = e^{2\pi i t}$ is not path homotopic to a constant path (w.l.o.g. $(1,0)$).

Assume there is a path homotopy $F : I \times I \to X$ such that $F(x,0) = h(x)$ and $F(x,1) = (1,0)$. W.l.o.g. we may assume $F : I \times I \to \mathbb{S}^1$: if F' is a path homotopy, then $F := \frac{x}{|x|} \circ F'$ is one too, because $\frac{x}{|x|}$ is continuous on X.

From here on we use the covering map (which we will define later) $p : \mathbb{R} \to \mathbb{S}^1$, $p(t) = e^{2\pi i t}$. By a lemma which we will prove later, F has a lift $\tilde{F}$, i.e. a continuous map $\tilde{F} : I \times I \to \mathbb{R}$ with $p \circ \tilde{F} = F$. We can normalize $\tilde{F}$ such that $\tilde{F}(0,0) = 0$.

Because $F(0,t) = F(1,t) = F(x,1) = (1,0)$ are all mapped to 1 (we use $\mathbb{R}^2 \cong \mathbb{C}$ for convenience) and $p^{-1}(1) = \mathbb{Z}$, the maps $\tilde{F}(0,t)$, $\tilde{F}(1,t)$ and $\tilde{F}(x,1)$ are continuous with values in $\mathbb{Z}$, and $\mathbb{Z}$ is discrete. Hence each of them is constant. Since consecutive edges share a corner, the three constants agree, and we get

$$\tilde{F}(0,0) = \tilde{F}(0,1) = \tilde{F}(1,1) = \tilde{F}(1,0) = 0.$$

We have now determined every edge except the bottom one. On the bottom edge, both $x \mapsto \tilde{F}(x,0)$ and $x \mapsto x$ are continuous lifts of h. We see that

$$p\big(\tilde{F}(x,0) - x\big) = \frac{h(x)}{h(x)} = 1,$$

so the difference lies in $\mathbb{Z}$. Since the difference is continuous on the connected set I, it is constant. At $x = 0$ the difference is 0, hence $\tilde{F}(x,0) = x$. But this is a contradiction, because $\tilde{F}(1,0) = 0 \neq 1$.

Definition 2.5.
Let f be a path in X from x_0 to x_1 and g be a path in X from x_1 to x_2. Then the concatenation of f and g is the path $f * g : I \to X$ defined by

$$f * g(x) = \begin{cases} f(2x) & 0 \le x \le \frac{1}{2} \\ g(2x-1) & \frac{1}{2} \le x \le 1 \end{cases}$$

This induces a product on the path homotopy equivalence classes: $[f]*[g] := [f*g]$.

Theorem 2.6.

(i) $[f] * ([g] * [h]) = ([f] * [g]) * [h]$

(ii) *Let $x \in X$ and e_x be the constant path at x. Let f be a path from x to y. Then $[e_x] * [f] = [f]$ and $[f] * [e_y] = [f]$.*

(iii) Let f be a path from x_0 to y. Then there exists an inverse path $\bar{f}$ from y to x_0 defined by $\bar{f}(x) = f(1-x)$. Then $[f] * [\bar{f}] = [e_{x_0}]$ and $[\bar{f}] * [f] = [e_y]$.

Proof.

(i) $[f] * ([g] * [h]) = [f * (g * h)] = [(f * g) * h] = ([f] * [g]) * [h]$ where the second equality follows from a reparametrization of the homotopy.

(ii) Let e_0 be the constant path at I and $id = i$ on I. We see that $i \simeq e_0 * i$. Then $f = f \circ i \simeq f \circ (e_0 * i) = (f \circ e_0) * (f \circ i) = e_{x_0} * f$. A similar argument shows $[f] * [e_y] = [f]$.

(iii) Let f be a path from x_0 to y. Then $\bar{f}(x) = f(1-x)$ is a path from y to x_0. It follows that $(f * \bar{f}(x)) = \begin{cases} f(2x) & 0 \le x \le \frac{1}{2} \\ f(2-2x) & \frac{1}{2} \le x \le 1 \end{cases} = f(t|1-2|x-\frac{1}{2}||)$. Let $F : I \times I \to X$ be defined by $F(x,t) = f(t|1-2|x-\frac{1}{2}||)$. Then $F(x,0) = f(0)$ and $F(x,1) = f(|1-2|x-\frac{1}{2}||)$. It follows that $(f * \bar{f}) \simeq_F e_{x_0}$. A similar argument shows $[\bar{f}] * [f] = [e_y]$.

□

Definition 2.7.
A **loop** based at $x_0 \in X$ is a path $f : I \to X$ with $f(0) = f(1) = x_0$.
$\pi_1(X, x_0)$ is the so called **fundamental group of X relative to** $x_0 (= \{[f] : f$ is a loop based at $x_0\})$. This is a group.

Example 2.8.
$\pi_1(\mathbb{R}^n, x_0) = 0$ for all $x_0 \in \mathbb{R}^n$. Also, $\pi_1(X, x_0) = 0$ for all $x_0 \in X$ if $X \subseteq \mathbb{R}^n$ is convex.

Question: Does $\pi_1(X, x_0)$ depend on the choice of x_0?

Definition 2.9.
Let f be a loop based at x_0. We define $\alpha : [0,1] \to X$ as a path from x_0 to x_1. It is easy to see that $\bar{\alpha} * f * \alpha$ is a loop based at x_1.
We define a map $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $\hat{\alpha}([f]) = [\bar{\alpha} * f * \alpha]$.

Theorem 2.10.
$\hat{\alpha}$ is a group isomorphism.

Proof.
Let f, g be loops based at x_0. Then $\hat{\alpha}([f] * [g]) = \hat{\alpha}([f * g]) = [\bar{\alpha} * (f * g) * \alpha] = [\bar{\alpha} * f * e_{x_1} * g * \alpha] = [\bar{\alpha} * f * (\alpha * \bar{\alpha}) * g * \alpha] = [\bar{\alpha} * f * \alpha] * [\bar{\alpha} * g * \alpha] = \hat{\alpha}([f]) * \hat{\alpha}([g])$.
If $\beta = \bar{\alpha}$, then $\hat{\beta} = \hat{\alpha}^{-1}$. Because $\hat{\alpha}$ is a homomorphism and it has an inverse, it is a group isomorphism. □

Corollary 2.11.
Let X be path connected. Then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$ for all $x_0, x_1 \in X$.

Definition 2.12.
X is called simply connected if it is path connected and $\pi_1(X, x_0) = 0$ for some (and hence all) $x_0 \in X$.

Theorem 2.13.
Let $h : (X, x_0) \to (Y, y_0)$ be a continuous map with $h(x_0) = y_0$. Then h induces a homomorphism $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ with $h_*([f]) = [h \circ f]$ for all $[f] \in \pi_1(X, x_0)$.

Proof.
h_* is well defined: if $f \simeq g$ then $h \circ f \simeq h \circ g$.
h_* is a homomorphism: let f, g be loops based at x_0. Then $h_*([f] * [g]) = h_*([f * g]) = [h \circ (f * g)] = [h \circ f * h \circ g] = [h \circ f] * [h \circ g] = h_*([f]) * h_*([g])$. □

Theorem 2.14.
If h is a homeomorphism then h_* is an isomorphism.

Proof.
Let $k = h^{-1}$. Then $k_* \circ h_* = (k \circ h)_* = id_* = id$. □